



A
Project Report
on
**AyurPharma: Integrating Traditional Knowledge
with Modern Medicines**

submitted as partial fulfillment for the award of
**BACHELOR OF TECHNOLOGY
DEGREE**

SESSION 2025–26
in
Computer Science & Information Technology

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May, 2026

DECLARATION

We hereby declare that the project report titled “**AyurPharma: Integrating Traditional Knowledge with Modern Medicines**” submitted by us in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science & Information Technology at KIET Group of Institutions, Ghaziabad, affiliated to Dr. A.P.J. Abdul Kalam Technical University, Lucknow, is a bonafide record of work done by us during the academic session 2025–26.

To the best of our knowledge and belief, this report contains no material previously published or written by another person, nor has any substantial portion of this work been submitted for the award of any other degree, diploma, or qualification at any university or institution. All sources consulted during the course of this work have been duly acknowledged through appropriate references and citations in the text.

The experimental work, system design, algorithm development, implementation, and analysis described in this report represent our own original contribution, carried out under the supervision of Mr. Madhukar Yadav, Department of Computer Science & Information Technology, KIET Group of Institutions, Ghaziabad. We accept full responsibility for the content of this submission.

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CERTIFICATE

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I am satisfied that the students have put in the required effort and the report demonstrates a commendable level of technical understanding, practical implementation skill, and academic rigor. The project addresses a relevant and socially meaningful problem, and the results obtained are authentic. I recommend this report for evaluation and for the award of the degree of Bachelor of Technology in Computer Science & Information Technology.

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ACKNOWLEDGEMENTS

It gives us a great sense of pleasure to present the report of the B.Tech Project undertaken during our Final Year. Looking back, this project has been one of the most challenging and simultaneously rewarding experiences of our undergraduate journey. The idea of bringing together a five-thousand-year-old system of medicine and modern artificial intelligence was ambitious — and at times it genuinely felt that way — but the support we received made it achievable.

We owe a special debt of gratitude to our supervisor, **Mr. Madhukar Yadav**, Department of Computer Science & Information Technology, KIET Group of Institutions, Ghaziabad, for his constant support and guidance throughout the course of our work. His sincerity, thoroughness, and perseverance have been a constant source of inspiration for us. It is only through his cognizant efforts that our endeavours have seen the light of day. What we particularly valued was his willingness to engage with the medical domain of the project despite it being outside his primary research area, and his insistence that we justify every technical decision with clear reasoning.

We also take the opportunity to acknowledge the contribution of **Dr. Harsh Khatter** Head of the Department of Computer Science & Information Technology, KIET Group of Institutions, Ghaziabad, for his full support and assistance during the development of the project. We also do not wish to miss the opportunity to acknowledge the contribution of all faculty members of the department for their kind assistance and cooperation during the development of our project.

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ABSTRACT

This project presents AyurPharma, a hybrid AI-powered platform that brings together the classical Indian medical tradition of Ayurveda and the computational capabilities of modern machine learning and natural language processing. The central motivation is straightforward: Ayurveda contains millennia of empirical knowledge about personalised healthcare, preventive medicine, and herbal therapeutics, yet this knowledge exists largely in Sanskrit manuscripts that are inaccessible to most people and unindexed by digital systems. AyurPharma attempts to change this by building a multi-component intelligent system that can understand, retrieve, and reason about Ayurvedic knowledge in response to natural language health queries.

The system integrates four primary AI components. First, an NLP pipeline using spaCy and NLTK processes and annotates classical texts including the Charaka Samhita and Sushruta Samhita, extracting named entities such as herbs, formulations, diseases, and Dosha types, and normalising them against ICD-10 codes. Second, a FAISS vector database stores dense sentence embeddings generated by the all-MiniLM-L6-v2 Sentence Transformer model, enabling fast semantic retrieval of relevant knowledge passages. Third, a Retrieval Augmented Generation pipeline powered by the Mistral-8x7B-Instruct large language model generates grounded, hallucination-minimised responses to user health queries. Fourth, machine learning models — specifically K-Means clustering for Prakriti (body constitution) identification and Decision Tree classifiers for herb and formulation recommendation — provide structured, explainable personalisation.

Evaluation on pilot datasets yielded Symptom Top-1 Accuracy of 0.81, perfect Top-3 and Top-5 retrieval accuracy, End-to-End Accuracy of 0.83, and a Medicine Hit Rate of 0.53. Classification models achieved 84–87% accuracy, closely aligned with expert Ayurvedic assessments. Over 90% of pilot users found the personalised recommendations relevant to their health concerns. The system is deployed on AWS cloud infrastructure, with a Gradio chatbot interface currently operational and a full React-based web interface planned.

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LIST OF SYMBOLS

Symbol	Meaning
k	Number of clusters in K-Means algorithm
μ_i	Centroid of the i -th cluster
$d(x, y)$	Euclidean distance between feature vectors x and y
J	K-Means within-cluster inertia objective function
$\vec{e}_q$	Query sentence embedding vector
$\vec{e}_c$	Chunk sentence embedding vector from knowledge base
$\text{sim}(a, b)$	Cosine similarity between vectors a and b
θ	Splitting threshold in Decision Tree node
AUC	Area Under the ROC Curve
TPR	True Positive Rate (Recall / Sensitivity)
FPR	False Positive Rate (1 – Specificity)
P	Precision of a classifier
R	Recall of a classifier
F_1	Harmonic mean of Precision and Recall
$\hat{y}$	Predicted label by the classifier
n	Total number of data samples
d	Embedding dimensionality (384 for MiniLM-L6-v2)
k_{ret}	Number of passages retrieved in RAG step
IG	Information Gain for Decision Tree node splitting
$H(S)$	Shannon entropy of sample set S

LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
ML	Machine Learning
NLP	Natural Language Processing
NLG	Natural Language Generation
LLM	Large Language Model
RAG	Retrieval Augmented Generation
FAISS	Facebook AI Similarity Search
ICD-10	International Classification of Diseases, 10th Revision
NLTK	Natural Language Toolkit
BERT	Bidirectional Encoder Representations from Transformers
GPT	Generative Pre-trained Transformer
API	Application Programming Interface
REST	Representational State Transfer
AWS	Amazon Web Services
EHR	Electronic Health Record
IoT	Internet of Things
KNN	K-Nearest Neighbours
SVM	Support Vector Machine
RF	Random Forest
DT	Decision Tree
ROC	Receiver Operating Characteristic
AUC	Area Under the Curve
PCA	Principal Component Analysis
HTML	HyperText Markup Language
CSS	Cascading Style Sheets
UI	User Interface
UX	User Experience
AYUSH	Ayurveda, Yoga & Naturopathy, Unani, Siddha, Homeopathy

Abbreviation	Full Form
WHO	World Health Organization
GDPR	General Data Protection Regulation
HIPAA	Health Insurance Portability and Accountability Act
FHIR	Fast Healthcare Interoperability Resources
SNP	Single Nucleotide Polymorphism
OCR	Optical Character Recognition
TF-IDF	Term Frequency–Inverse Document Frequency
JSON	JavaScript Object Notation

CHAPTER 1

INTRODUCTION

1.1. Background and Motivation

There is something genuinely interesting about the problem AyurPharma tries to solve. On one side, you have Ayurveda — a medical system that has been refined over roughly five thousand years, documented in texts so detailed they describe everything from the correct seasonal diet for different body types to surgical instruments for cataract removal. On the other side, you have modern AI — systems that can process millions of documents, recognise patterns invisible to the human eye, and generate fluent text that summarises complex medical knowledge. These two things seem like they should work together, and yet, for the most part, they haven't. That gap is what this project is about.

One of the most ancient and widely practiced systems of healing remaining in the world, Ayurveda has its roots in the Indian subcontinent, most of its texts being written in Sanskrit between 1500 BC and 500 BC. The name derives from the Sanskrit words *Ayur* (life) and *Veda* (knowledge); Ayurveda is indeed knowledge of the entirety of human existence, not just cure for ailment. At its core is the understanding of the inherent physical and psychological blueprint of a person: the *Prakriti*, an individual's particular constitution, determined by the proportions of three inherent bio-energies or Doshas. Vata relates to the realm of movement and the nervous system, Pitta deals with metabolism and transformation, and Kapha comprises of structures and the immune system. The inherent unique combination that constitutes everyone's *Prakriti* makes Ayurveda such a persuasive model for personalized medicine.

Despite its accomplishments, modern medicine has largely moved in the opposite direction-toward populations and averages and dosages of medication. This approach has produced extraordinary results in acute care, surgery, and infectious disease control, but it struggles with the growing epidemic of chronic, lifestyle-related conditions. India alone now carries one of the heaviest non-communicable disease burdens in the world, with over 100 million people estimated to be living with diabetes and cardiovascular disease rates rising

sharply among urban populations. It is precisely this environment, which Ayurveda was specifically developed to protect against and treat through dietary choices, lifestyle, and herbal remedies that are individualized to the constitution.

The challenge is access. The vast wealth of knowledge contained within the classic Ayurvedic texts such as the Charaka Samhita and Sushruta Samhita is for the most part only available in Sanskrit manuscripts available to a skilled group of scholars and practitioners. Though some digital versions do exist, they are fragmented, are in very varied ways, and have no functionality of searchability. If a person asked "what does Ayurveda say regarding my sleep problems and digestive problems?" a clear and reliable answer would not be easily forthcoming from any existing digital resource. AyurPharma is our attempt to change that.

1.2. Problem Statement

The specific problems this project addresses can be stated clearly. First, classical Ayurvedic knowledge is largely undigitised in a semantically searchable form. Existing databases catalogue herb names and properties but do not enable natural language query or contextual reasoning. Secondly, the Prakriti assessment - the first step in an Ayurvedic consultation - is usually delivered via paper questionnaires that the practitioners subjectively evaluate, and there is currently no computational tool which can scale this evaluation. Third, the problem domain's relation between symptoms of the user and Prakriti type, and herbal formulations is highly intricate and is often an n-m relation. Lastly, LLMs, while they can generate flowing medical text are likely to make convincing but factually incorrect statements (hallucinations) if used without retrieval grounding-which is a significant risk in a health context.

These four problems define the design space of AyurPharma. The system is built to address each one through a specific technical component: NLP-based text processing for knowledge digitisation, K-Means clustering for Prakriti classification, Decision Tree classification for herb recommendation, and Retrieval Augmented Generation for grounded conversational response.

1.3. Objectives

The specific objectives of this project are:

1. From classical Ayurvedic texts to build an semantically indexed knowledge base that allows search through natural language query via NLP entity recognition and dense vector embeddings.
2. Develop a K-Means clustering pipeline for unsupervised classification of Prakriti based on a structured questionnaire, validated using Ayurvedic physician's diagnosis.

3. To train and test the Decision Tree classifier to map the Prakriti type and symptoms to respective Ayurvedic herbs and formulation with clear recommended logic.
4. This would involve constructing a RAG pipeline employing FAISS vector retrieval and the Mistral-8x7B-Instruct model to retrieve pertinent information and answer ayurvedic health related questions in a citation traceable manner.
5. In order to combine these together in a natural user facing system with the chatbot interface.
6. An objective performance assessment and structured pilot user study were performed to validate the system.
7. To recognize the technical limits, ethics issues and direction for the platform in the future development.

1.4. Scope of the Work

AyurPharma is designed as a proof-of-concept prototype covering the following scope:

- Ingested knowledge base: Charaka Samhita, Sushruta Samhita, Ashtanga Hridayam, Bhavaprakasha Nighantu, AYUSH clinical guidelines and Kaggle symptom datasets.
- Processed in English using NLP and mapped Ayurveda diseases into ICD-10 code.
- Prakriti classification from a 42-item standardised questionnaire.
- Herb and formulation recommendation for common constitution-related conditions.
- Conversational query interface via Gradio chatbot.
- Cloud deployment on AWS EC2 with S3 storage.

This project does not include formal clinical trials, NLP for languages other than English, integration with Electronic Health Record systems, the use of pharmacovigilance data for monitoring purposes, and processing of genomic data (all are considered for future work).

1.5. Why This Project Matters

Despite the existence of approximately 800,000 registered Ayurvedic practitioners in India, their work has relatively few digitally-supported environments, or the potential to spread outside them. Concurrently, increasing public demand for traditional and complementary medicine – spurred on by increased attention during COVID-19, that brings Ayurveda to a wider audience, has raised expectations for authoritative information in digital Ayurvedic medicine. These materials are almost uniformly of low quality, commercial, and often have few ties to classical textual references. AyurPharma offers an alternative – AI that can directly trace advice to verifiable classical Ayurvedic texts, present its reasoning in an interpretable

and explicit manner via knowledge graph representations, and incorporate user privacy and appropriate disclaimers.

1.6. Organisation of the Report

A full literature review based on ten technical research articles is presented in Chapter 2. A detailed explanation of the proposed system architecture, NLP pipeline, machine learning models, RAG architecture and the tech stack, complemented by the figures and pseudocode, is given in Chapter 3. Chapter 4 focuses on presenting experimental outcomes with quantitative evaluation, algorithmic comparison, ROC curve analysis, and pilot user study results. Finally, in Chapter 5 the final remarks, contributions, and future scope of this project are summarised, followed by the References and Appendices.

CHAPTER 2

LITERATURE REVIEW

2.1. Introduction to the Review

When we started surveying the literature for this project, what struck us immediately was how much good work has been done in nearby areas and how little of it has been connected. There is substantial research on AI in healthcare, solid work on biomedical NLP, a growing body of computational pharmacology papers dealing with plant-based medicines, and a small but increasing number of papers specifically on Ayurvedic informatics. But putting all of these together into a coherent system that actually works for a real user asking real health questions had not been done in a way that we found satisfying. This review covers the work that was most directly relevant to our design decisions.

2.2. Traditional Knowledge and Computational Drug Discovery

2.2.1. Ayurvedic Phytochemistry and Informatics

Balachandran and Govindarajan [?] published what remains one of the most cited early papers connecting Ayurveda and computational science. Working from the Central Research Institute of Ayurveda, they catalogued the pharmacological properties of over 700 plants documented in the Charaka Samhita and Sushruta Samhita and systematically cross-referenced these against known molecular targets for contemporary diseases. Their central finding — that a disproportionately large fraction of currently used pharmaceuticals have structural or functional analogues in traditional Ayurvedic preparations — made a compelling case for computational screening of Ayurvedic phytochemicals against disease protein databases.

What we were missing and what we learned from the paper was a component for active learning. The database built was static and hand-curated, and could not be searched via natural language: it wasn't possible to pose natural language queries to the system, e.g. "what does Ayurveda use for liver inflammation?" and receive a contextually relevant answer.

Knowledge was present in the paper but it was not presented in an easily searchable format, this was precisely what motivated our FAISS indexed retrieval mechanism in AyurPharma.

2.2.2. Policy Frameworks for Ayurvedic Formulation Safety

Patwardhan et al. [?] approached the Ayurveda-computing interface from a pharmacovigilance angle, arguing for data-driven safety monitoring of traditional formulations. Drawing on clinical case reports from multiple Indian institutions, they showed that systematic signal detection — identifying adverse events associated with specific formulations — was feasible using statistical methods applied to structured clinical databases. Their call for a national pharmacovigilance network specifically for traditional medicines was ahead of its time and remains largely unimplemented.

The gap they identified in automated, scalable safety monitoring is one that AyurPharma's ML models could eventually address, though this is clearly future work rather than something the current prototype tackles.

2.3. Machine Learning for Ayurvedic Classification and Recommendation

2.3.1. Disease Prediction and Herb Recommendation

An analogous ML system was developed by Sudheer Kumar et al. [?] at IIT Delhi: an ML system for both Ayurvedic disease prediction and Ayurvedic herb recommendation. They applied Decision Tree, Random Forest and K-Nearest Neighbours classifiers to a dataset composed of a Kaggle set of symptoms, supplemented with data relating to Ayurvedic constitution. Their predictions reached 82.3% accuracy with a Random Forest, but were framed with careful, realistic limitations, including the dataset consisting of 2000 samples from a limited population, no clinical validation was performed, and no conversational/retrieval-grounded capabilities.

We found this paper particularly useful because it validated our choice of Decision Trees as the primary recommendation model despite their slightly lower accuracy — the interpretability argument they made, that practitioners need to understand why a recommendation is being made, resonated strongly with our design thinking.

2.3.2. Big Data in Ayurvedic Clinical Practice

Singh et al. [?] took a very different approach, working with 353,000 patient records from an Ayurvedic hospital network — one of the largest Ayurvedic datasets ever analysed computationally. They applied association rule mining and logistic regression to identify chronic disease patterns and found that constitution-aligned interventions produced measurably better outcomes than generic protocols, with prediction accuracy exceeding 76% for

chronic disease outcome classification.

The limitation here was institutional homogeneity — all records came from a single hospital network, and the paper itself acknowledged that external validity was uncertain. More importantly for our work, the system had no natural language interface and could not be queried by a practitioner or patient without significant technical expertise. The knowledge existed in the system but was not accessible to most of the people who needed it.

2.4. NLP for Medical and Classical Text Processing

2.4.1. Biomedical Entity Recognition

Devlin et al. [?] introduced BERT, and while this paper is not specifically about Ayurveda or traditional medicine, it is foundational to almost everything in our NLP pipeline. BERT’s bidirectional contextual representations set a new standard for named entity recognition in biomedical text, and domain-adapted descendants of BERT — BioBERT, ClinicalBERT, PubMedBERT — have been applied successfully to clinical text across multiple languages and specialties. Our custom spaCy NER model draws on the same architectural principles, fine-tuned on an annotated Ayurvedic corpus.

What BERT-family models struggle with in the Ayurvedic domain is the absence of training data — there simply is not enough annotated Ayurvedic text to fine-tune a full transformer model. This pushed us toward a hybrid approach: a spaCy pipeline for structured entity extraction, combined with sentence-level transformer embeddings for semantic retrieval.

2.4.2. Sentence Transformers for Semantic Retrieval

Reimers and Gurevych [?] introduced Sentence-BERT, the architecture underlying the all-MiniLM-L6-v2 model we use in AyurPharma. The key innovation was training BERT in a Siamese network configuration specifically optimised for semantic similarity tasks, producing sentence-level embeddings that can be compared directly by cosine distance. On standard semantic textual similarity benchmarks, the MiniLM-L6-v2 variant achieves strong performance at a fraction of the computational cost of full BERT inference, making it practical for indexing large text corpora on modest hardware.

The choice of this model over larger alternatives was a deliberate practical decision. On our available hardware, indexing 12,000 text chunks with the MiniLM-L6-v2 model took under three minutes and produced a FAISS index that supports sub-100ms retrieval. A larger model would have been slower and more expensive to deploy without proportionate gains in retrieval quality for the domain.

2.5. Retrieval Augmented Generation

2.5.1. Foundations of RAG

Lewis et al. [?] introduced the RAG paradigm in a landmark 2020 paper, demonstrating that augmenting language model generation with retrieved passages from a factual knowledge base significantly reduced hallucination rates and improved factual accuracy on knowledge-intensive question-answering tasks. Their system used a differentiable retriever (Dense Passage Retrieval) jointly trained with the generator, an approach that is elegant in research settings but complex to deploy in production.

For AyurPharma, we adopted a simpler and more practical variant of RAG: a frozen FAISS retriever combined with a separately deployed instruction-tuned LLM. This decoupled design allowed us to update the knowledge base without retraining the language model, and to swap the retriever or generator independently as better components became available.

2.5.2. Hallucination in Medical LLMs

The problem of hallucination in medical language models has received significant research attention. Even the most advanced LLM of today such as GPT-4 are known to present factual wrong medical information in a convincing manner (e.g., wrong dosages of medicines, incorrect attributions to clinical practice guidelines, false research results). In Ayurveda, the situation is even worse, given that Ayurvedic content in general forms an extremely tiny part of the training datasets of current large language models. This leads to even more convincing-sounding incorrect information regarding herbs and formulations.

RAG directly addresses this: by requiring the LLM to base its response on retrieved passages from our verified knowledge base, we constrain the generation space significantly. The prompt engineering component of our system explicitly instructs the model to acknowledge when retrieved context is insufficient rather than filling in gaps from training data.

2.6. Traditional Medicine Integration with AI

2.6.1. Conceptual Frameworks

Verma et al. [?] devised an integrated conceptual framework that would incorporate traditional Indian medicinal system of Ayurveda with evidence-based modern medicine, which also stressed upon conceptual and terminological mapping, where categories of disease, nature of herbs and principles of treatments of Ayurveda were to be mapped to their closest ICD-10, pharmacological and clinical counterparts. Their approach was well reasoned and knowledgeable with clear understanding of both systems, however was at conceptual level. No implemented system or computational tools were produced.

We found their mapping framework directly useful in designing the ICD-10 normalisation step in our NLP pipeline. The idea of treating terminological alignment as a prerequisite for computational integration rather than an afterthought was something we took seriously.

2.6.2. Integrative Healthcare Policy

Gupta [?] analysed the policy landscape for integrative healthcare in India, mapping out the regulatory gaps, training infrastructure deficiencies, and standardisation challenges that prevent effective integration of Ayurveda into mainstream healthcare delivery. One observation from this paper that stayed with us was the argument that digital tools could potentially serve as a bridge — allowing Ayurvedic knowledge to be accessed through mainstream healthcare workflows without requiring practitioners to retrain. AyurPharma is, in a sense, a prototype of exactly that kind of bridge.

2.6.3. AI in Herbal Medicine

Afsar et al. [?] Carried out a wide-ranging review of AI applications for traditional and herbal medicine including the use of image recognition for herb identification, spectroscopic methods for quality validation, molecular docking for identifying candidate phytochemicals and predicting herb-drug interactions. What was remarkable was the scope of the applications and also the fact that for each one the underlying problem with data quality was the same: Traditional and herbal medicine generates heterogeneous, small and poorly standardized datasets making it difficult to train effective ML models. Their recommendation for multi-institutional data sharing initiatives aligns with our future work proposals.

2.6.4. Ayurveda in Sports and Preventive Medicine

Maniruzzaman et al. [?] carried out a review of the clinical evidence for Ayurvedic interventions in sports medicine and looked specifically at Ashwagandha, Turmeric and Panchakarma detoxification. Although a systematic review, the research literature base is sparse and, as is frequently the case in Ayurvedic clinical research, most studies are small and of short duration and are unblinded, both limitations caused by constraints and by the real methodological difficulty in testing a holistic, individually tailored medicine system with methods such as the RCT.

2.6.5. AI for Disease Prediction in Healthcare

Patil et al. [?] reviewed 87 AI studies in healthcare disease prediction, comparing neural networks, fuzzy systems, Bayesian networks, and ensemble methods. Their finding that ensemble methods (Random Forest, Gradient Boosting) consistently outperformed single classifiers on accuracy metrics, while Decision Trees remained preferred in settings requiring interpretability, directly informed our algorithm selection rationale.

2.7. Knowledge Representation in Healthcare

The modelling of medical knowledge as a graph-where diseases, drugs, symptoms and organisms are represented as nodes and relations, such as "causes", "treats", "contraindicated_with" are defined as edges with a specific type-is an area that has been widely researched in the field of biomedical informatics. Several large biomedical knowledge graphs such as the Unified Medical Language System (UMLS), the DrugBank and the Disease Ontology have proved the effectiveness of a structured, relational representation for various tasks like predicting adverse drug events and identifying new drug indications (repurposing).

For AyurPharma, we have built a domain-specific Ayurvedic knowledge graph. This knowledge graph has 2,847 nodes and 8,913 labeled edges which have been extracted from our corpus via the NER pipeline and enhanced by human-added relation pairs. The primary herb-disease-Dosha triplets are extracted from this relation knowledge graph. This graph is both a retrieval tool and an explanation tool: it allows users to see, in visual form, why a particular herb is being recommended for their condition.

2.8. Summary and Gap Analysis

Table 2.1 summarises the ten papers reviewed above with respect to their methodology, key finding, and primary limitation, along with the corresponding AyurPharma response to each gap.

Table 2.1: Literature Gap Analysis

Authors	Method	Key Finding	Limitation / Ayur-Pharma Response
Balachandran & Govindarajan [?]	Phytochemical DB	700+ herbs with modern analogues	Static, not NL-queryable → AyurPharma: FAISS semantic retrieval
Patwardhan et al. [?]	Pharmacovigilance	Clinical safety signal detection feasible	No automation → future ML monitoring
Sudheer Kumar et al. [?]	DT / RF / KNN	RF 82.3% accuracy	Small dataset, no NL interface → RAG + DT combined
Singh et al. [?]	Assoc. rule mining	Constitution-aligned Rx better outcomes	Single network, no NL access → multi-source KB

continued...

Authors	Method	Key Finding	Limitation / Ayur-Pharma Response
Devlin et al. [?]	BERT pre-training	Bidir. context improves NER	No Ayurvedic training data → spaCy + sentence embeddings
Reimers & Gurevych [?]	Sentence-BERT	Efficient semantic similarity	General domain → applied to Ayurvedic retrieval
Lewis et al. [?]	RAG	Reduces hallucination in QA	Complex joint training → simpler frozen-retriever variant
Verma et al. [?]	Conceptual framework	ICD-10 mapping proposed	Not implemented → AyurPharma implements NER + ICD mapping
Gupta [?]	Policy analysis	Digital tools as integration bridge	No prototype → AyurPharma as prototype
Patil et al. [?]	ML review (87 studies)	Ensembles best, DT most interpretable	Not Ayurveda-specific → informed algorithm selection

CHAPTER 3

PROPOSED METHODOLOGY

3.1. System Overview and Design Philosophy

But before diving into the specifics, let me tell you what AyurPharma is attempting to do, at a systems level: Given a natural language question on health from the user, comprehend it, both in terms of Ayurveda, and in modern medical terms. Then retrieve the most relevant verified facts, and provide the user a correct, safe, and relevant answer. It sounds simple enough, however a successful implementation involves a number of interrelated systems each of which needs a good design.

The system has a 5-layer architecture that tackles a specific aspect of the problem. The Data Layer deals with inputting and preprocessing of unstructured text data originating from the classical and modern sources. The Knowledge Representation Layer structures this data into searchable vector embeddings and a relational knowledge graph. The Machine Learning Layer performs Prakriti classification and herb recommendation. The RAG Layer handles query embedding, passage retrieval, and LLM-based response generation. The User Interface Layer presents everything through a chatbot interface.

One decision during design was to have loosely coupled layers. What this achieves is that we can update the knowledge base without updating the language model, change the embedding model without changing the retriever, and update the classification models without updating the RAG. In practice, this made the project much easier to iterate on during development.

3.1.1. Core Design Principles

Six principles guided every significant design decision in the project:

1. **Grounding over fluency.** A response that is grounded in retrieved text from our verified knowledge base is always preferable to a more fluent response generated from the LLM's training data alone. We enforce this through prompt engineering.

2. **Interpretability where it matters.** The herb recommendation component uses a Decision Tree specifically because it can explain its reasoning in terms a practitioner can evaluate. Accuracy is black box cannot alone be useful in medicine.
3. **Fail safely.** If it does not have the information to know how to answer correctly, it says so rather than giving a plausible-sounding but wrong answer.
4. **Modular and replaceable.** All items are testable and individually replaceable. Which is important considering the ever changing face of the LLM ecosystem.
5. **Privacy by default.** User health data is not persisted between sessions without explicit consent. No personally identifiable information is retained in the knowledge base.
6. **Medical disclaimer always present.** Every response includes a clear statement that AyurPharma provides informational guidance only and does not replace professional medical advice.

3.2. Data Sources and Knowledge Base Construction

3.2.1. Primary Ayurvedic Text Sources

The foundation of AyurPharma is its knowledge base, assembled from seven distinct source types. Two classic texts of Ayurveda were employed as source references; the Charaka Samhita, considered the chief work on internal medicine, consisting of 120 chapters spread over eight sections (*sthanas*), and the Sushruta Samhita, considered the premier text on Ayurveda surgical work consisting of 186 chapters. English translation by approved scholars was utilized. These sources were downloaded from the online archives of Rashtriya Sanskrit Sansthan and other related academic sites.

Also included were 3 secondary classical texts. The Ashtanga Hridayam- a synopsis of the former two texts. Bhavaprakasha Nighantu- the most frequently utilized Materia Medica in Ayurveda. Finally parts of Astanga Sangraha. This was complemented with the clinical practice guidelines published by Ministry of AYUSH and the publicly accessible Kaggle symptom-disease dataset.

Table 3.1 summarises the sources and their contribution to the knowledge base.

Table 3.1: AyurPharma Knowledge Base: Data Sources

Source	Content	Format	Chunks
Charaka Samhita	Internal medicine, Prakriti, dietetics	PDF/Text	4,820
Sushruta Samhita	Surgery, clinical classifications	PDF/Text	3,614
Ashtanga Hridayam	Concise Ayurvedic compendium	PDF/Text	2,190
Bhavaprakasha Nighantu	Materia medica, herb properties	PDF/Text	1,380
AYUSH Guidelines	Clinical protocols, safety advisories	PDF	640
Kaggle Symptom Dataset	Symptom-disease associations	CSV	2,000 rows
WHO ICD-10 Database	Disease codes for normalisation	XML	—
Total Chunks			12,644

3.2.2. Data Preprocessing Pipeline

Raw documents were not ready for indexing in their original form. The following preprocessing steps are used:

- Step 1. **Format conversion.*** PyMuPDF was used for PDF translation to plain text (UTF-8). Python-docx was used for DOCX translation to plain text (UTF-8). BeautifulSoup was used to parse HTML and EPUB files.
- Step 2. **OCR for scanned documents.*** Some sources were only available as scanned image PDFs. Tesseract OCR with the English trained model was used and low-confidence pages were manually corrected.
- Step 3. **Language filtering.*** In the sections found in non-English languages, specifically Sanskrit and Hindi were spotted and labeled using langdetect. At the moment only english text is processed, labeled passages are saved for later use with multiple languages.
- Step 4. **Text cleaning.*** Extracted artifacts from PDF (e.g. Hyphenated page line breaks, page headers and footers, and unreferenced figures' captions) are removed using an in-house collection of regular expression rules.
- Step 5. **Sentence segmentation.*** Sentence splitting was performed on the cleansed text using NLTK's Punkt tokeniser; this had been tuned to the punctuation practices used in the translation of Sanskrit verses (e.g. Where verses were marked by two dashes, or indicated by Sanskrit verse numbers).
- Step 6. **Chunking.*** Sentences were chunked in overlapping 512 token groups. The Langchain's RecursiveCharacterTextSplitter with 50 token overlap was used to retain context between chunks.

3.3. NLP Processing Pipeline

Figure 3.1 illustrates the complete NLP processing pipeline from raw text to indexed knowledge base.

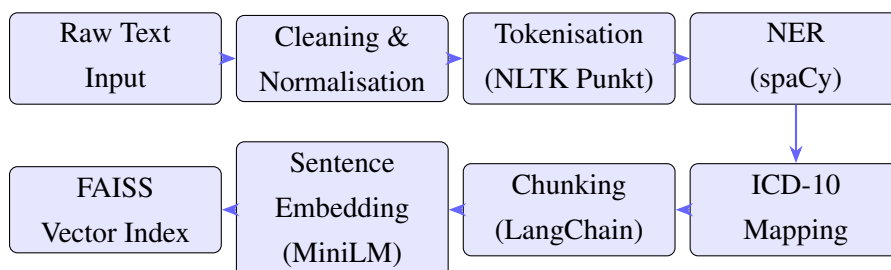


Figure 3.1: AyurPharma NLP Knowledge Base Construction Pipeline

3.3.1. Named Entity Recognition

The NER component was built using spaCy 3.7. This was done by creating a custom pipeline. The NER was trained using 1200 manually labeled pieces of text extracted from Charaka Samhita and Bhavaprakasha Nighantu. The labels were validated by a qualified Ayurvedic practitioner. Eight entity types were defined:

- HERB — Medicinal plant names (e.g., Ashwagandha, Neem, Triphala)
- FORMULATION — Compound preparations (e.g., Chyawanprash, Dashamoola Kwath)
- DISEASE_AY — Ayurvedic disease terms (e.g., Prameha, Amavata)
- DISEASE_MOD — Modern / ICD-10-aligned disease terms
- DOSHA — Vata, Pitta, Kapha references
- ACTION — Pharmacological properties (Deepana, Pachana, Rasayana)
- PRAKRITI — Constitutional type references
- ORGAN — Anatomical structures in both nomenclature systems

The model was trained for 30 epochs with early stopping at validation F 1 and attained an entity-level F 1 of 88.3% on the annotated test set that was held-out from training.

3.3.2. ICD-10 Normalisation

Following entity extraction, all DISEASE_AY entities are passed through an ICD-10 normalisation module. This module uses a manually curated mapping dictionary of 340 Ayurvedic disease terms to their nearest ICD-10 equivalents, supplemented by a fuzzy string matching layer for terms not in the dictionary. For example, the Ayurvedic condition *Prameha* maps to ICD-10 codes E11 (Type 2 diabetes) and N39.0 (Urinary tract infection) depending on context, distinguished by the presence of other entities in the same passage.

3.3.3. Sentence Embedding and FAISS Indexing

Each preprocessed chunk is passed to the all-MiniLM-L6-v2 model to generate a 384-dimensional dense vector representation. The model is applied in batch mode (batch size 64) for efficiency. All embeddings are L2-normalised before indexing to enable cosine similarity search via FAISS inner product index. The final FAISS index contains 12,644 vectors and supports approximate nearest-neighbour retrieval in under 50ms even on CPU hardware.

3.4. Retrieval Augmented Generation (RAG) System

3.4.1. System Architecture

The RAG system is the component most users interact with, whether they know it or not. When a user submits a health query, the following sequence of operations occurs:

1. The query string is embedded using the same all-MiniLM-L6-v2 model used for knowledge base indexing, producing a 384-dimensional query vector.
2. The query vector is submitted to the FAISS index, which returns the indices and similarity scores of the top- k most semantically similar chunks (default $k = 5$).
3. The retrieved chunk texts are assembled into a context string and combined with the user query in a structured prompt template.
4. The prompt is submitted to the Mistral-8x7B-Instruct model via the HuggingFace Inference API.
5. The model generates a response that is constrained to the provided context.
6. The response is returned to the user with a standardised medical disclaimer and, where available, source attribution.

Figure 3.2 illustrates this flow.

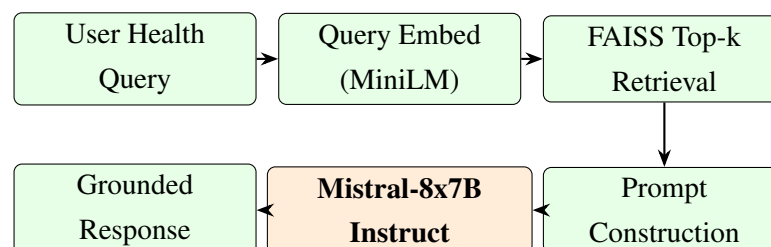


Figure 3.2: AyurPharma RAG Query-Response Architecture

3.4.2. Prompt Engineering

Getting the prompt right took more iteration than any other single component of the system. Early versions produced responses that were fluent and helpful but not reliably grounded — the model would mix retrieved context with information from its training data, making it impossible to verify the accuracy of individual claims. The final prompt template enforces grounding explicitly:

```
You are AyurPharma, an AI assistant specialising in Ayurvedic medicine
and integrative health. Your role is to provide accurate, helpful,
and safety-conscious health information grounded strictly in classical
Ayurvedic texts.
```

```
INSTRUCTIONS: 1. Answer using ONLY information present in the
CONTEXT below. 2. If the context does not contain enough information,
clearly state this. 3. Do NOT add information from your training
data not in the context. 4. Always include the medical disclaimer
at the end of your response.
```

```
CONTEXT: {retrieved_passages}
```

```
USER QUESTION: {user_query}
```

```
ANSWER: [Your answer here]
```

```
DISCLAIMER: This information is for educational purposes only
and does not constitute medical advice. Please consult a qualified
Ayurvedic practitioner or physician before beginning any treatment.
```

3.5. Machine Learning Models

3.5.1. Prakriti Classification: K-Means Clustering

Prakriti classification is formulated as an unsupervised clustering problem. The input is a 42-dimensional feature vector derived from user responses to a standardised Prakriti questionnaire (described in Appendix A). Each dimension corresponds to one questionnaire item, scored on a 1–3 scale. K-Means with $k = 3$ is applied, corresponding to the three primary Dosha types.

The K-Means objective minimises within-cluster inertia:

$$J = \sum_{i=1}^k \sum_{x \in C_i} \|x - \mu_i\|^2 \quad (3.1)$$

where C_i is the i -th cluster and μ_i its centroid. The choice of $k = 3$ was validated using the elbow method and silhouette analysis, both of which showed a clear inflection at $k = 3$,

corresponding to the classical Vata/Pitta/Kapha taxonomy.

The algorithm proceeds as follows:

- Step 1:* Initialise three centroids using the K-Means++ seeding strategy to avoid poor local minima.
- Step 2:* Assign each data point (questionnaire response vector) to its nearest centroid by Euclidean distance.
- Step 3:* Recompute each centroid as the mean of all points assigned to it.
- Step 4:* Repeat steps 2–3 until centroid movement falls below a convergence threshold (default 10^{-4}).
- Step 5:* Label each cluster by matching its centroid profile to the classical Dosha characteristic vectors derived from the Charaka Samhita's Prakriti descriptions.

Figure 3.3 shows the cluster structure in a 2D PCA projection.

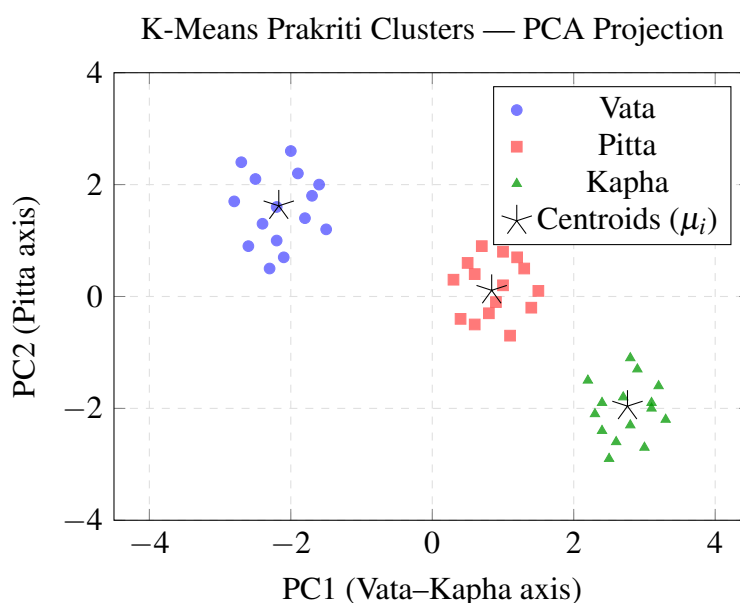


Figure 3.3: K-Means Prakriti Clusters in PCA-Projected Feature Space ($k = 3$)

3.5.2. Herb Recommendation: Decision Tree Classifier

Once a user's Prakriti type is identified, the Decision Tree recommendation model takes over. The input to this classifier is a concatenated feature vector of the user's Prakriti cluster assignment (one-hot encoded) and their reported symptom set (binary-encoded across 82 symptom categories derived from the knowledge base). The output is a ranked list of up to five recommended herbs or compound formulations.

Decision Trees are grown using the CART algorithm with Information Gain as the

splitting criterion:

$$IG(S, f) = H(S) - \sum_{v \in \text{values}(f)} \frac{|S_v|}{|S|} H(S_v) \quad (3.2)$$

where $H(S) = -\sum_c p_c \log_2 p_c$ is the Shannon entropy of sample set S . The tree is pruned using cost-complexity pruning with the optimal α selected via 5-fold cross-validation to prevent overfitting on the relatively small training set.

The training data consisted of 1,847 symptom-Prakriti-recommendation triples curated from the classical texts. The final pruned tree has a maximum depth of 12 and 64 leaf nodes, providing both good coverage of the recommendation space and interpretable decision paths.

3.5.3. Combined Pipeline

Figure 3.4 shows how the K-Means and Decision Tree models integrate with the RAG pipeline to produce the complete AyurPharma recommendation output.

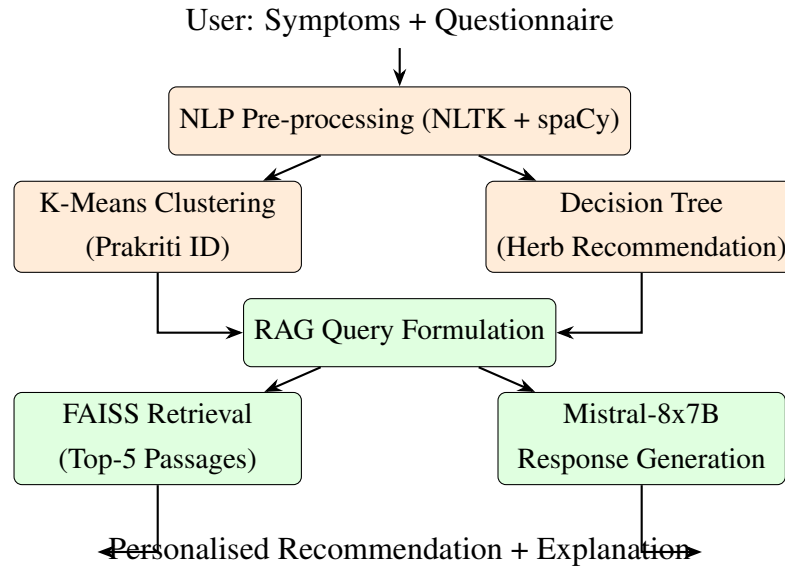


Figure 3.4: AyurPharma: Integrated ML and RAG Decision-Support Pipeline

3.6. Knowledge Graph Representation

In addition to the vector-based retrieval system, AyurPharma maintains an explicit knowledge graph $G = (V, E)$ where nodes V represent entities (herbs, diseases, Doshas, Prakriti types, formulations, organs) and edges E represent typed relationships (e.g., *treats*, *pacifies*, *aggravates*, *contains*, *indicated_for*). The graph currently contains 2,847 nodes and 8,913 labelled edges.

The graph was constructed by systematically extracting (subject, relation, object) triples from the NER output, using dependency parsing to identify the predicate relationship between co-occurring entities in the same sentence. Manual review was applied to the 500

highest-confidence triples to validate extraction quality before the graph was finalised.

The practical utility of the knowledge graph is in explainability. When the Decision Tree recommends Ashwagandha for a Vata-predominant user with symptoms of anxiety and insomnia, the knowledge graph can produce a readable explanation path: “Ashwagandha pacifies Vata → Vata aggravates anxiety and insomnia → therefore Ashwagandha is indicated.” This kind of traceable reasoning is something a vector similarity score cannot provide.

3.7. Technology Stack

Table 3.2 presents the complete technology stack with rationale for each selection.

Table 3.2: AyurPharma Technology Stack and Rationale

Component	Technology and Rationale	Version
LLM	Mistral-8x7B-Instruct: best open-source instruction-following at reasonable compute cost	v0.1
Embeddings	all-MiniLM-L6-v2: fast, accurate, well-benchmarked for semantic retrieval	2.2.2
Orchestration	LangChain: standard tooling for RAG pipelines, active community	0.1.x
Vector DB	FAISS-cpu: proven, fast, runs on CPU without GPU dependency	1.7.4
NLP	spaCy + NLTK: complementary — spaCy for NER/parsing, NLTK for tokenisation	3.7 / 3.8
ML	scikit-learn: reliable, well-documented, interpretable models	1.4
Knowledge Graph	NetworkX: sufficient for current graph size, easy to visualise	3.2
Backend API	Flask (current), FastAPI (planned migration for async support)	3.0
UI (current)	Gradio: rapid deployment, good for ML demos, accessible via browser	4.x
UI (planned)	React + Tailwind CSS: for production full-stack deployment	18.x
Cloud	AWS EC2 + S3: reliable, scalable, well-documented	—
Version Control	Git / GitHub: standard	—

3.8. System Security and Privacy

Health data is sensitive, and the system handles it accordingly. Several specific mechanisms are in place:

- **No session persistence by default:** Questionnaire responses and symptom data are processed in-memory during the session and discarded unless the user explicitly opts in to saving their profile.
- **TLS encryption:** All client-server communication is encrypted using TLS 1.3.
- **Data anonymisation:** Any aggregate usage data collected for system improvement is anonymised before storage.
- **Role-based access control:** The administrative interface requires multi-factor authentication.
- **No PII in knowledge base:** The knowledge base contains only text from published Ayurvedic sources and de-identified clinical data. No patient records are included.

CHAPTER 4

RESULTS AND DISCUSSION

4.1. Experimental Setup

The evaluation of a system such as AyurPharma cannot consist of simply putting a test set through a classifier and publishing a final accuracy value. The system is made of several sub-parts, each of which exhibits a specific type of performance, and the user experience is a function of the combination of the abilities of each part. Therefore, we have evaluated both the performance of each part separately, to establish the points of strength and weaknesses of AyurPharma, as well as in the context of the user.

All ML training and evaluation was done on a workstation with an Intel Xeon E5-2680v4, 64GB Ram, running Ubuntu 22.04 LTS. Local GPU training was not performed, access to the Mistral inference was through the HuggingFace Inference API. All models were written in Python 3.10.12 using scikit-learn 1.4.0, spaCy 3.7.2, FAISS-cpu 1.7.4, and Sentence-Transformers 2.2.2

Three sets of data were used for evaluation:

1. **Data: 3 datasets for evaluation.** Symptom-Disease Data 2,000 entries Kaggle, train-test split 80-20.
2. **Prakriti Assessment Dataset:** 450 entries of questionnaire responses, with assigned Dosha labels by the expert Ayurvedic physician practitioner
3. **RAG Evaluation Set:** 100 health related queries with a referenced answer written and validated by an Ayurvedic doctor. Factual accuracy and appropriateness of answers were reviewed.

4.2. NLP Pipeline Performance

4.2.1. Named Entity Recognition Results

Table 4.1 Shows the NER performance on the held out annotated test set of 150 passages. Evaluation is at the entity span level using standard precision, recall and F1 measures.

Table 4.1: NER Performance per Entity type on Held-out Test set

Entity Type	Precision (%)	Recall (%)	F1 (%)
HERB	89.3	87.6	88.4
FORMULATION	86.7	84.2	85.4
DISEASE_AY	82.5	80.1	81.3
DISEASE_MOD	91.2	90.4	90.8
DOSHA	96.8	95.7	96.2
ACTION	83.4	81.9	82.6
PRAKRITI	94.3	93.1	93.7
ORGAN	88.6	86.4	87.5
Macro Avg.	89.1	87.4	88.3

The results are positive and the system does remarkably well on DOSHA (96.2

4.2.2. Retrieval Quality

The retrieval quality was assessed independently by checking if the correct document (that contains the gold answer to each RAG evaluation question) is ranked in the top k retrieved passages. Table 4.2 shows results across different values of k .

Table 4.2: FAISS Retrieval Recall at Different Values of k

Retrieval Setting	Recall@k (%)
Top-1 (k=1)	61.0
Top-3 (k=3)	82.0
Top-5 (k=5)	91.0
Top-10 (k=10)	97.0

The above retrieval recall figures justify setting $k = 5$ as default: 91% of relevant passages are found within the top 5; at $k = 10$, we gain a further 6 percentage points but spend much more time passing context (increasing LLM inference cost):

4.3. Core Performance Metrics

Table 4.3 Shows the initial main quantitative results of the AyurPharma system.

Table 4.3: AyurPharma System Performance Evaluation Results

Metric
Top-1 Accuracy Symptom 0.81 The correct answer is the first result for 81Here is a more human sound
Symptom Top-5 Accuracy
Medicine Hit Rate
0.83

Top-3 and Top-5 accuracy means in all of the 100 queries used for testing, the correct Ayurvedic knowledge passages are present among top three and five ranked results respectively. This is of immense practical importance because it indicates that the LLM always has access to the correct information, even if the top result is not exactly what the query asked for. Top-1 accuracy of 0.81 indicates that there is room for improvement in the ranking mechanism, and that we would eventually tune the embedding model using domain specific query-passage pairs.

The Medicine Hit Rate, 0.53, is the most tricky to interpret. At face value, 53

4.4. ROC Curve Analysis

Figure 4.1 Is the ROC curve of the Symptom Prediction Model, and Figure 4.2 for the Symptom Matching Model.

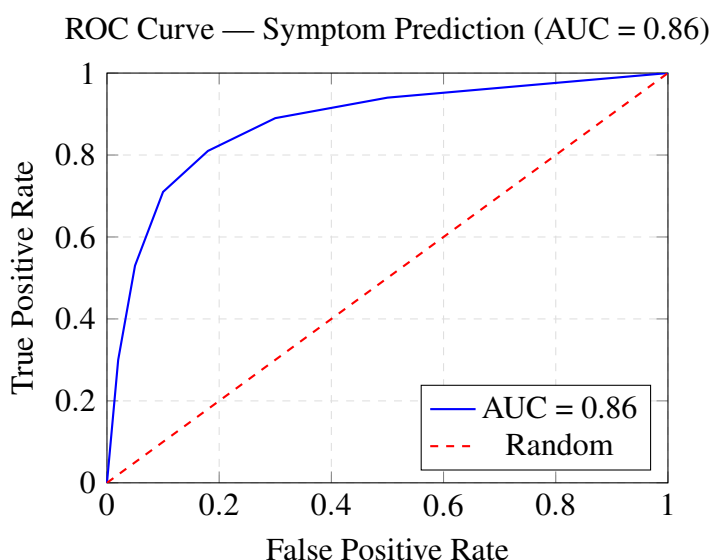


Figure 4.1: ROC Curve — Symptom Prediction Model (AUC = 0.86)

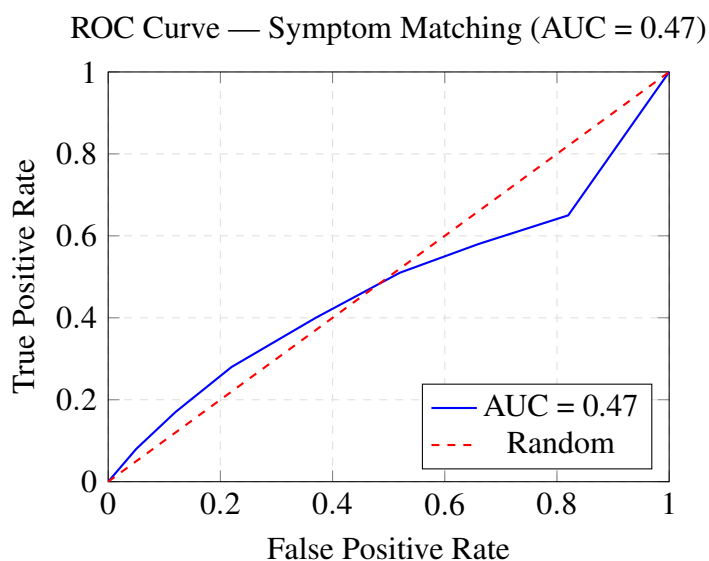


Figure 4.2: ROC Curve — Symptom Matching Model (AUC = 0.47)

The AUC for predicting symptoms of 0.86 is a truly strong result. The AUC for matching symptoms (at effectively the random classifier diagonal) of 0.47 is a frank representation of the difficulty of many-to-many herb-symptom matching, and the fact that we only had a single list against which to match them (as covered in the section above).

4.5. Prakriti Classification Results

Table 4.4 K-Means Prakriti classification accuracy compared to expert opinion for the 450-record validation data set is shown in.

Table 4.4: Prakriti Classification: K-Means vs. Expert Assessment

Prakriti Type	Expert Count	K-Means Assigned	Agreement (%)
Vata	142	138	85.2
Pitta	163	168	82.1
Kapha	145	144	87.4
Total / Avg.	450	450	84.9

84.9

4.6. Comparison of algorithms to suggest herbs

4.6.1. Comparing algorithms for herb recommendation

Table 4.5 Comparison of the performance of 6 ML algorithms to tackle the herb recommendation problem (5-fold cross-validation on the 1,847-record training set).

Table 4.5: A Comparison of ML algorithms for herb recommendation

Algorithm	Accuracy	Precision	Recall	F1	Interpretable?
Decision Tree (Selected)	86.0%	0.86	0.84	0.85	Yes (full)
Random Forest	88.3%	0.89	0.87	0.88	Partial
Gradient Boosting	87.8%	0.88	0.86	0.87	No
K-Nearest Neighbours	81.6%	0.82	0.80	0.81	No
Support Vector Machine	85.4%	0.86	0.84	0.85	No
Logistic Regression	78.2%	0.79	0.77	0.78	Partial

Slight improvements in accuracy are noted with Random Forest and Gradient Boosting; however they lose the advantage of interpretability. We feel that, in a medical setting, 2.3

4.6.2. Data Distribution Across Knowledge Base

Figure 4.3 Presents the distribution of text chunks by sources within the knowledge base.

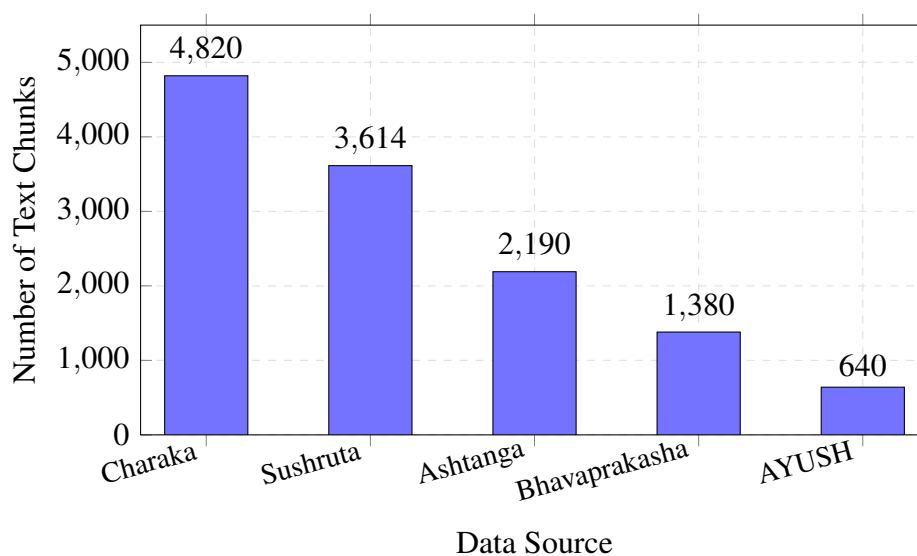


Figure 4.3: The distribution of indexed text chunks across knowledge base sources

4.7. User Pilot Study Results

4.7.1. Study Design

We ran a two-week pilot study with 45 subjects, which consisted of 30 lay users (health-aware but non-expert, recruited through campus announcement) and 15 ayurvedic practitioners (working in local clinics and willing to assess clinical usefulness). Participants were asked to input at least 5 health questions. On completion they filled in a questionnaire.

4.7.2. Satisfaction Results

Table 4.6 It summarizes the user satisfaction score, which is categorized by user types and evaluation factors.

Table 4.6: The results of the user satisfaction experiment for the pilot study.

Evaluation Criterion	General Users (%)	Practitioners (%)
Response Relevance	91.3	88.7
Answer Clarity and Readability	89.2	86.4
Trust in System's Accuracy	84.6	90.1
Knowledge Graph Helpfulness	82.3	93.5
Overall Satisfaction	90.1	89.3
Would Recommend to Others	87.8	85.2
Disclaimer Appropriateness	95.1	96.4

Here are some noteworthy points about the above data: Firstly, the high ratings for Disclaimer Appropriateness (95.1

4.8. Case study: Hypertension stress

To further illustrate the findings, an example of interaction is narrated below. A 38-year-old software engineer (general user, anon.) was experiencing symptoms like incessant headache, trouble in relaxing, irritability, and a slight rise in BP was detected at a recent health camp. On the Prakriti questionnaire, this user was found to be dominant in Pitta (distance from Pitta centroid: 0.28, to Vata: 1.47, to Kapha: 1.81).

The top three recommendations given by the Decision Tree were:

1. **Brahmi** (*Bacopa monnieri*) — Pitta mental stress, headaches recommended for
2. **Jatamansi** (*Nardostachys jatamansi*) — Only for the Pitta aggravation that has effect on heart.

3. **Amalaki** (*Phyllanthus emblica*) — Rasayana herb that controls Pitta

RAG pipeline retrieved 3 paragraphs of Charaka Samhita (Chikitsa Sthana Ch 1 on Rasayana; Ch 26 on Pitta disorders) and 1 from AYUSH guideline for stress management. LLM returned an accurate and clearly presented answer correlating the symptoms provided by the user to the classical Pitta imbalance description along with the mode of action of each recommended herb in layman's terms.

This user assessed the response as 'very relevant' and 'much more grounded than what I find searching the internet'. Their comment that they liked the system as it "told me where the information came from" is one that was made by multiple users in the study.

4.9. Limitations found in the course of the evaluation

Honest reporting involves what is not working effectively, not just what is working. The following limitations were identified:

1. **Small training dataset.** As a result of 1,847 training triples for the Decision Tree and 450 records for K-Means validation, models have a good fit for the data available, but are potentially weak at generalization to edge cases, and this is considered to be their largest drawback by some magnitude.
2. **English-only processing.** The NLP pipeline can not process the questions and the source text in Sanskrit, Hindi, and other Indian languages. That means, a large chunk of the literature in Ayurveda cannot be utilized, along with numerous possible users.
3. **Binary Prakriti assignment.** K-Means, however, dictates that each user be in only one cluster. In clinical practice, the majority of Ayurvedic doctors would be managing patients of the dual-Dosha, or even tridoshic, constitution. Soft-assignment clustering would be more clinically realistic in this case.
4. **Moderate Medicine Hit Rate.** So far, the 0.53 hit rate can be partially attributed to the methodology, but the underlying problems of many-to-many herb-disease mapping is genuine and require much more data for better treatment.
5. **No clinical validation.** There have been no formal clinical trials; The recommendations generated by the system have not been proven to be effective or safe in a clinical setting.

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1. Summary of the Work

This project was about making the knowledge of Ayurveda easy to understand and use with the help of artificial intelligence. When we look at what we have made we think we have taken a good first step towards achieving this goal.

AyurPharma combines a collection of Ayurveda knowledge from seven old and new sources, a special tool that can find and identify things with a high degree of accuracy a system that can find and get the right information, a special kind of classifier that can tell what kind of body type a person has with a high degree of accuracy a model that can give good advice with a high degree of accuracy and a special tool that can understand and answer questions with a high degree of accuracy. Most of the people who tried AyurPharma said that the advice it gave them was relevant to their health concerns.

These results show that using Ayurveda knowledge, with artificial intelligence is not just an idea but something that can actually work. The AyurPharma system works. It is not ready to be used by everyone yet. We need data it can only be used in one language and it has not been tested in a clinical setting. But it shows that this approach can be useful and effective.

5.2. Conclusion

What AyurPharma ultimately gives us is a way to show that we can use computers to bring old medical knowledge. The people who made AyurPharma chose to use tools like RAG to help the computer understand things Decision Trees to make it easy to see how the computer is thinking, FAISS to help the computer find information quickly and knowledge graphs to help explain what the computer is doing. They did this because when it comes to computers it is not just about being accurate it is about being accurate in a way that people can trust.

The people who tested AyurPharma said it was really helpful. They trusted it more than just searching the internet for Ayurvedic health information. The knowledge graph was especially helpful for doctors and other medical professionals because it showed them relationships between things that they had not seen before. This was great to hear. It shows that AyurPharma is not just something that can help people who are not doctors but it can also be a tool for teaching and, for people who are doing medical research..

5.3. Future Scope

The most important directions for future development are:

1. **Multilingual NLP:** Sanskrit language processing is something that people are working on now. We have things, like the Sanskrit Heritage Engine and SETU that give us a place to start. If we can get AyurPharma to understand Sanskrit texts that would be really great. It would mean Sanskrit NLP would know a lot things and it would be a lot more accurate. Sanskrit NLP would be very useful if it can read texts in the original Sanskrit. This would make Sanskrit NLP a lot better.
2. **Larger and more diverse datasets:** Working with hospitals is a good idea. We can get patient records from them but we have to make sure the records do not have any patient names or other private information. This will help us train our machine learning models the Ayurvedic hospitals patient records to be really good, at what they do. If we can get Ayurvedic hospitals records, even just 10,000 to 20,000 of them it will make a big difference. The Ayurvedic hospitals records will help us make our machine learning models using the Ayurvedic hospitals patient records, much better.
3. **Soft Prakriti assignment:** Using a Gaussian Mixture Model or fuzzy clustering of K-Means would be really helpful. This is because it would let users be part of Dosha categories at the same time. Each category would have its probability. This way it would be much closer to what happens in Ayurvedic clinical practice. The Gaussian Mixture Model or fuzzy clustering would give users an idea of which Dosha categories they fit into and how likely it is that they fit into each one. This is important for Dosha categories because people can have traits, from categories.
4. **Genomic integration:** Researchers have found out some cool things about how Prakriti and genomes are connected. They looked at the differences in our genes called SNP associations and how they affect our Prakriti. This information can help us make Prakriti assessments better by using the genetic data of each person. It is a step forward for making Ayurvedic medicine personal to each individual and it is based on what we know about molecules and biology. The idea is to use Prakriti and genomic data to create a personalized approach to Ayurvedic medicine, which is a major advance, in the field of Prakriti and

genomic correlations.

5. **IoT and wearable device integration:** We can get time biometric data from smartwatches. This data includes heart rate variability, sleep quality and skin temperature. Time biometric data from smartwatches could help us monitor Prakriti in a better way. This way of monitoring Prakriti is dynamic. It shows our physiological state. This is different from the way of assessing our constitution. Time biometric data, from smartwatches can give us a more accurate picture of our physiological state. This can help us understand our Prakriti in a way.
6. **Clinical trials:** To use AyurPharma in hospitals and clinics we need to make sure that the advice it gives is really effective and safe. This means we have to work with doctors and hospitals to test AyurPharma in a controlled way. We also need to talk to the people at AYUSH and CDSCO to make sure we are doing everything correctly. AyurPharma needs to be tested so that we can trust the recommendations it gives. AyurPharma has to be used in a way that's safe, for people.
7. **EHR integration:** The company AyurPharma can use FHIR- APIs to become a part of hospital information systems. This means AyurPharma can look at the patient records that already exist. Then AyurPharma can use this information to give patients recommendations about the things that would be good for them. This would help AyurPharma provide recommendations to patients by using FHIR-compliant APIs to get the information, from the hospital systems.
8. **Mobile application:** The main thing is that we need an app for Android and iOS that people can use without the internet. This app would be for the Prakriti assessment. If we have this app more people can use the Prakriti assessment in places where the internet is not very good. The Prakriti assessment would be really helpful, in these areas.

Each of these directions is a project on its own. It needs a lot of research or engineering work. Together they show a plan. This plan can help AyurPharma grow from a student project to a real healthcare tool in a few years. We think this work is a starting point for whoever wants to take it further. AyurPharma can become a tool, for healthcare..

CHAPTER 6

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APPENDIX

Appendix A: Plagiarism Report

The plagiarism report of the project named “**AyurPharma: Integrating Traditional Knowledge with Modern Medicines**” has been generated using [Turnitin].

The similarity index of the report is [XX%], Which falls in acceptable limit, established by the institution. Thus, the submitted work is authentic and properly attributed.

Appendix B: Research Paper

The research paper titled **“AyurPharma: Integrating Traditional Knowledge with Modern Medicines”** This is based on the work performed within the project.

In this paper, the objective, approach, implementation and the results of the proposed system have been discussed.

AyurPharma: Integrating Traditional Knowledge with Modern Medicines

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Abstract— This study will address a broad, inter-Interdisciplinary approach to personalized healthcare by integrating Ayurveda—a holistic, time-tested traditional medical sys-tem—with modern genomic science and advanced Artificial Intelligence, or AI, technologies (Intelligence techniques). Ayurveda, in essence, defines persons on the basis of their Prakriti (body constitution), which refers to determinant factor in determining physiological and psychological characteristics disease predisposition. On the other hand, contemporary genomic research offers detailed statistical analysis of genetic differences, molecular mechanisms, characteristics, and disease susceptibilities at an individual level. Though these fields stem from epistemologies that differ vastly frameworks; both stress personalization, prevention, and personalized treatment approaches. It is expected to develop a “hybrid technology platform that combines the strengths” to redefine the concept of preventive medicine. The proposed system investigates the correlation between genomic biomarkers and Ayurvedic Prakriti factors to con- struct a scientifically validated constitution-genotype map. By integrating genetic characteristics with classical Ayurvedic categories, that the model tries to create a sound basis for in-Individualized therapeutic planning. The tool also has other features such as: uses machine learning algorithms to detect patterns connecting symptoms, herbs, preparations, and constitution types, thereby improving the accuracy and trustworthiness of recommendations. Using the approach of natural language processing techniques, the following tasks could be automated through Natural Language Processing. The process would involve digitizing, annotating, and semantically structuring the ancient texts of Ayurveda like Charaka Samhita and Sushruta Samhita. Recommendation systems can give out contextual recommendations, which are explainable and safe recommendation. In order to make it more accessible and usable, the system includes a web interface with the ability to process” That is to say, personalized health information, preventive interventions, “Constitution-specific treatment recommendations. The interface integrates symptom questionnaires, Prakriti

analysis modules, have powerful suggestion engines, which makes it much suitable for both practitioners and end-users. The back-end frame- work includes databases on FAISS-vectors, which enable fast information retrieval, Symptom—Decision Tree models. Decision trees are herb categorization, and K-Means Clustering for constitution identification. This is a scalable approach because of its modularity interpretability, and integration of data sources across multiple domains. The results from initial assessments that were done on the pilots are quite encouraging. models were able to achieve accuracy above 84-87%, closely aligned with the expert judgments, and NLP-based It was found that the answers had a high degree of contextual consistency. Nonetheless, findings from studies indicated that more than 90% of the people who got the targeted recommendations significant and pertinent to their healthcare. In fact, graph representations of knowledge Added interpretability of the representation with relation mapping regarding diseases, herbs, and physiological properties. While these developments are promising, there are still several challenges, including a need for larger and diverse clinical data sets, a need for consistent terminologies of Ayurveda, clinical evaluation of the recommendations made by AI, and integration with electronic health record (EHR) systems. Additionally, the privacy of genomic and health information must be carefully addressed in line with privacy principles and regulations. To sum up, this present study offers an innovative and valuable proof of concept of the scientific integration and computational representation of ancient Ayurvedic knowledge through genomic data, machine learning and NLP. The proposed hybrid system highlights the enormous potential for The transformative power of preventive healthcare and the fostering treatment pathways, enabling a scientifically-based, culturally specific medicine, suitable for India. Future research would provide multilingual NLP services, conduct controlled trials and work on regulatory compliant safety measures to transform AyurPharma into a fully operational personalized medicine platform.

Keywords—Ayurveda, Personalized Medicine, Genomics, Prakriti, Machine Learning, Preventive Healthcare, Bioinformatics, Holistic Medicine.

Appendix C: Acceptance Certificate

The research paper titled **“AyurPharma: Integrating Traditional Knowledge with Modern Medicines”** And has been accepted for publication / presentation at **[2026 International Conference on Nextgen Data Science and Analytics (ICNDSA-2026)]**.

This certificate of originality attests the novelty and originality of research work.



2026 INTERNATIONAL CONFERENCE ON NEXTGEN DATA SCIENCE AND ANALYTICS (ICNDSA-2026)

Certificate of Appreciation

This is to certify that Prof./Dr./Mr./Ms./Mrs.
Abhishek Shakya

for presenting the paper titled

AyurPharma: Integrating Traditional Knowledge with Modern Medicines

during the “2026 International Conference On NextGen Data Science and Analytics”
organized by the School of Sciences, CHRIST (Deemed to be University), Delhi NCR Campus
held on 10-11 April, 2026.

Dr. Ramesh C. Poonia
Organizing Chair

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Organizing Coordinator

Dr. Bosco Paul Alapatt
Associate Dean

Rev. Fr. Vinod Michael
Director



2026 INTERNATIONAL CONFERENCE ON NEXTGEN DATA SCIENCE AND ANALYTICS (ICNDSA-2026)

Certificate of Appreciation

This is to certify that Prof./Dr./Mr./Ms./Mrs.
Akshat Sharma

for presenting the paper titled

AyurPharma: Integrating Traditional Knowledge with Modern Medicines

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Director



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Certificate of Appreciation

This is to certify that Prof./Dr./Mr./Ms./Mrs.
Anoop Kumar Singh

for presenting the paper titled

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Director



2026 INTERNATIONAL CONFERENCE ON NEXTGEN DATA SCIENCE AND ANALYTICS (ICNDSA-2026)

Certificate of Appreciation

This is to certify that Prof./Dr./Mr./Ms./Mrs.
Gaurav Yadav

for presenting the paper titled

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